

CS1231S Assignment #1

AY2023/24 Semester 2

Deadline: Wednesday, 21 February 2024, 1:00pm

IMPORTANT: Please read the instructions below.

This is a graded assignment worth 10% of your final grade. There are **six questions** (“question” 0 and six problem-solving questions) with a total score of 40 marks. Please work on it by yourself, not in a group or in collaboration with anybody. Anyone found committing plagiarism (submitting other’s work as your own), or sending your answers to others, or other forms of academic dishonesty will be penalised with a straight zero for the assignment, and possibly an F grade for the module. Please see SoC website “Preventing Plagiarism” <https://www.comp.nus.edu.sg/cug/plagiarism/>.

You are to submit your assignment to **Canvas > Assignments before the deadline.**

Your answers may be typed or handwritten. Make sure that it is legible (for example, don’t use very light pencil or ink if it is handwritten, or font size smaller than 11 if it is typed) or marks may be deducted.

You are to submit a **SINGLE pdf file**, where each page is A4 size. Do not submit files in other formats. If you submit multiple files, only the last submitted file will be graded.

Late submission will NOT be accepted. We will set the closing time of the submission folders to slightly later than 1pm to give you a few minutes of grace, but in your mind, you should treat **1pm** as the deadline. If you think you might be too busy on the day of the deadline, please submit earlier. Also, avoid submitting in the last minute; the system may get sluggish due to overload and you will miss the deadline.

Note the following as well:

- Name your pdf file with your **Student Number**. Your student number begins with ‘A’ (eg: A0234567X). (Do not mix up your student number with your NUSNET-id which begins with ‘e’.)
- At the top of the first page of your submission, write your **Name** and **Tutorial Group**.
- To keep the submitted file short, please submit your answers without including the questions.
- As this is an assignment given well ahead of time, we expect you to work on it early. You should submit **polished work**, not answers that are untidy or appear to have been done in a hurry, for example, with scribbling and cancellation all over the places. Marks may be deducted for untidy work.
- It is always good to be clear and leave no gap so that the marker does not need to make guesswork on your answer. When in doubt, the marker would usually not award the mark.
- Do not use any methods that have not been covered in class. When using a theorem or result that has appeared in class (lectures or tutorials), please quote the theorem number/name, the lecture and slide number, or the tutorial number and question number in that tutorial, failing which marks may be deducted. Remember to use numbering and give justification for important steps in your proof, or marks may be deducted.

To combine all pages into a single pdf document for submission, you may find the following scanning apps helpful if you intend to scan your handwritten answers:

* for Android: <https://fossbytes.com/best-android-scanner-apps/>

* for iphone:
<https://www.switchingtomac.com/tutorials/ios-tutorials/the-best-ios-scanner-apps-to-scan-documents-images/>

If you need any clarification about this assignment, please do NOT email us or post on telegram, but post on **QnA “Assignments”** topic so that everybody can read the answers to the queries.

Question 0. (Total: 2 marks)

Check that ...

- you have submitted a pdf file with your Student Number as the filename. [1 mark]
- you have written both your name and tutorial group number (eg: T02) at the top of the first page of your file. (If you miss either one you will not be given any mark.) [1 mark]

Question 1. Propositional logic (Total: 7 marks)

To prove part (a), show each step with justification using only Theorem 2.1.1 and the implication law (if necessary). To disprove, cite a counter-example and show that it is a correct counter-example.

For part (b), you may use appropriate rules of inference in addition to Theorem 2.1.1 and implication law.

Do not use truth table for both parts. Do not cite any other results that have not been shown in class.

(a) Prove or disprove: $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$ [3 marks]

(b) Aiken and Dueet want to invite three friends Xes, Yul, and Zam to their party but are given these constraints:

1. Xes comes only if Yul does not come.
2. Yul comes if Zam or Xes comes.
3. Xes comes if Zam does not come.

- (i) Express the above constraints in propositional logic. [1 mark]
- (ii) Who will come to the party and who will not? Show how you arrive at your conclusion by giving a logical argument. (Not by writing an essay!) [3 marks]

Question 2. Argument (Total: 7 marks)

(a) Determine the validity of the following argument: [2 marks]

“Zack does not have a keen sense of hearing. Zack is a pianist or a writer. Zack is a pianist only if he has a keen sense of hearing. Therefore, Zack is a writer.”

You may use the following statement variables:

p : Zack is a pianist.

w : Zack is a writer.

h : Zack has a keen sense of hearing.

(b) Given a particular tour group, the following argument is presented:

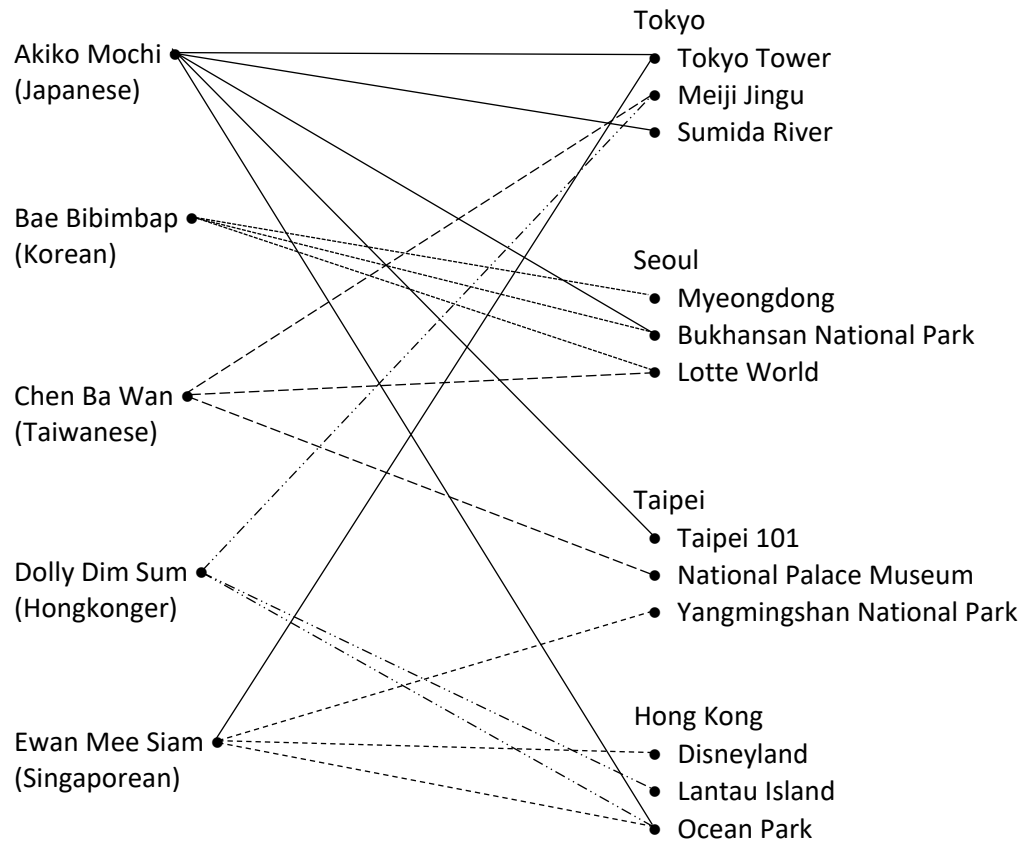
“A tourist in this tour group has not visited Gardens by the Bay. Every tourist in this tour group has visited Sentosa. Therefore, someone in this tour group who has visited Sentosa has not visited Gardens by the Bay.”

Define appropriate predicates and determine the validity of the above argument. (Write a proper proof, not an essay!) Let the domain of discourse be the set of all human beings and you may omit this set in your statements.

[5 marks]

Question 3. Quantified statements (Total: 8 marks)

- (a) The diagram below shows the places of interest in four cities where five visitors went to. You may assume that the Japanese is resident in Tokyo, the Korean in Seoul, and the Taiwanese in Taipei. In the context of this question, the places of interest listed are all the places of interest in that city. A line joining a visitor to a place of interest indicates that the visitor visited that place.



Answer the following parts with “true” or “false”, and give your explanation clearly. [5 marks]

- There is some visitor who visited all the places of interest in some city.
- Every visitor visited at least two places of interest in some city.
- Some of the places of interest in every city were visited solely by non-residents of that city.
- Every visitor visited some places of interest that were also visited by another visitor.
- The place of interest visited by the most visitors is located in the city visited by the most visitors.

- (b) Given the following predicates:

- $Cat(x)$: x is a cat.
- $Dog(x)$: x is a dog.
- $Socialist(x)$: x is a socialist.
- $Likes(x, y)$: x likes y .

Write a quantified statement for each of the following sentences. You may assume that the domain of discourse is the set of all animals A . [3 marks]

- No cat is a socialist.
- All cats are liked by some dog.
- No dog likes a socialist.

Question 4. (Quantified statements on sets) (5 marks)

Let $A = \{-1, 0, 1\}$, $B = \{0, 1, 4\}$ and $C = \{-2, -1, 0, 1, 2\}$. Indicate whether each of the following statements is true or false with explanation.

- (a) $\exists x \in A \forall y \in B \exists z \in C ((y \neq 0) \wedge (y > xz))$.
- (b) $\exists x \in A \forall y \in B \exists z \in C (z > xy)$.
- (c) $\forall x \in C ((\forall y \in B xy \in B) \rightarrow (x^2 = x))$.

Question 5. Sets (5 marks)

Prove or disprove the statements below. Do not use Venn diagrams in your proof or counter-example.

- (a) $\overline{A \setminus (B \cup C)} \subseteq \overline{A} \cap (B \cup C)$ for all sets $A, B, C \subseteq \mathbb{Z}$.
- (b) $(A \cup B) \setminus (C \setminus A) = A \cup (B \setminus C)$ for all sets A, B, C .

Question 6. Writing proof (6 marks)

Prove the following statement:

“If the square of an integer is divisible by 5, then that integer is divisible by 5.”

(Do not write proof that is not rigorous. For example, “if n^2 is divisible by 5, then n^2 ends with 0 or 5 (everybody knows this), and so n must also end with 0 or 5 (common sense).” Also do not use real number functions such as square root. Use only what have been covered in class.)

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